

PORTFOLIO

Gyeongbin Ryoo

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About me Skill/Career Projects

SKILL

Language	Web Technology	Infra & DB	Tool
Python C++ Java	HTML CSS	AWS MySQL	Visual Code Visual Studio Eclipse Git

Career

2021 (Freshman)

C++ Programming
Advanced Python Programming

2022 (Sophomore)

Advanced C++ Programming
AI Service Design

2023 (Junior)

Recommendation System
Big Data Analysis

2024 (Senior)

Capstone Design
Interdisciplinary Capstone Design

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Papers

1. "A Fine-Tuning Approach for Object Detection Models using Virtually Generated Data," accepted for presentation at ICTC 2024, 2024.10.17 (First co-author) , In Preparation for Publication [Download PDF]

Led research on fine-tuning and analyzing the performance of object detection AI models using virtual data collected from NVIDIA Omniverse, demonstrating that virtual data is effective in improving model performance in real-world environments

2. "Simulation-based Service Scenarios Analysis for Commercial Deployment: Robot-Friendly Building Case," accepted at the KOSIME Domestic Conference, 2024.07.05 (Co-author) [Download PDF]

Conducted research analyzing simulation-based scenarios for optimizing robot services based on the concept of "robot-friendly buildings," evaluating service delay times according to the number of robots through 3D simulations, and overcoming the limitations of existing methods

3. "Performance Analysis of Semantic Segmentation in Fine-tuned Models Based on Virtual Environment", presented at the Symposium of the Korean Institute of Communications and Information Sciences, 2024.06.19 (First author)" [Download PDF]

Conducted research utilizing data collected from a virtual environment to train an semantic segmentation model, followed by adding real-world data for performance comparison. The results demonstrated a significant improvement in the model's detection capabilities, confirming the usefulness of virtual data in real-world applications

4. "Analyzing Experimental Data of 6-DOF Motions and Sloshing Impacts in LNG Carriers," published at the ASME

International Conference on Ocean, Offshore & Arctic Engineering (OMAE2024), 2024.06.09 (Co-author)" [Download PDF]

Conducted research analyzing the relationship between the 6-DOF motions of LNG carriers and sloshing impact loads by utilizing experimental data from over 5,000 marine conditions. Using machine learning techniques such as LightGBM and GAM, the study demonstrated that acceleration is the most significant variable affecting sloshing loads

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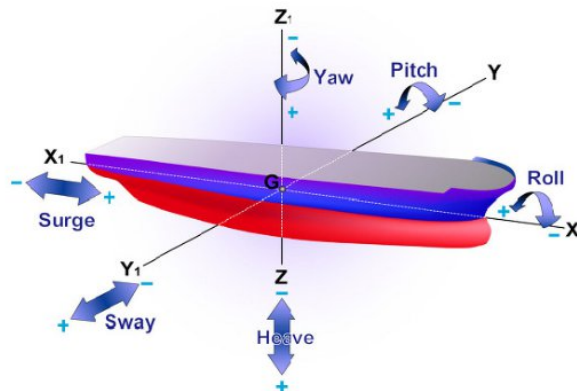
About me Skill/Career PROJECTS

Marine Sloshing AI Research (2023.03 ~ 2023.12)

Research Introduction:

Sloshing AI Prediction

Significant impacts occur due to sloshing during the transportation of LNG (liquefied natural gas), influenced by various factors such as the amount of liquid filled, the direction of the ship's movement, and the period and height of waves. To predict these sloshing impacts, we conducted research utilizing machine learning techniques.



▶ OA

Pytorch, Pandas, Numpy

▶ Language

Python, C/C++.

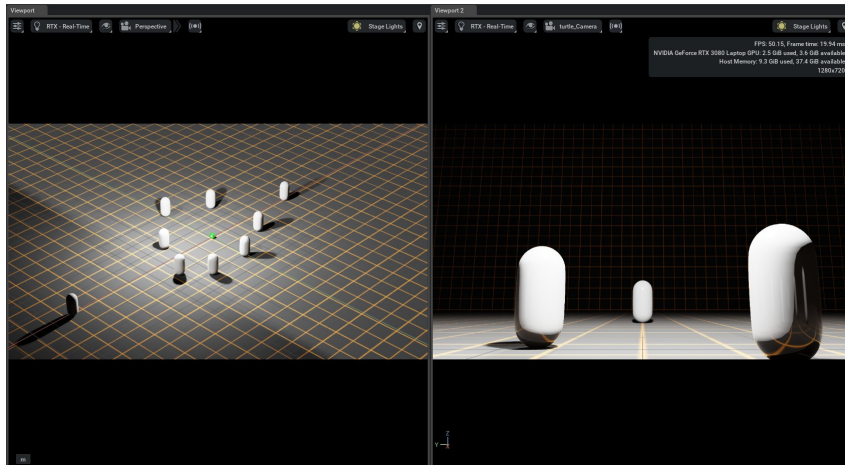
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Metaverse Research (NVIDIA Isaac Sim Platform) (2023.12 ~)

Research Introduction:

Metaverse Research Conducted research on extracting virtual environment data from the Nvidia Omniverse Isaac Sim platform and validating the significance of the data

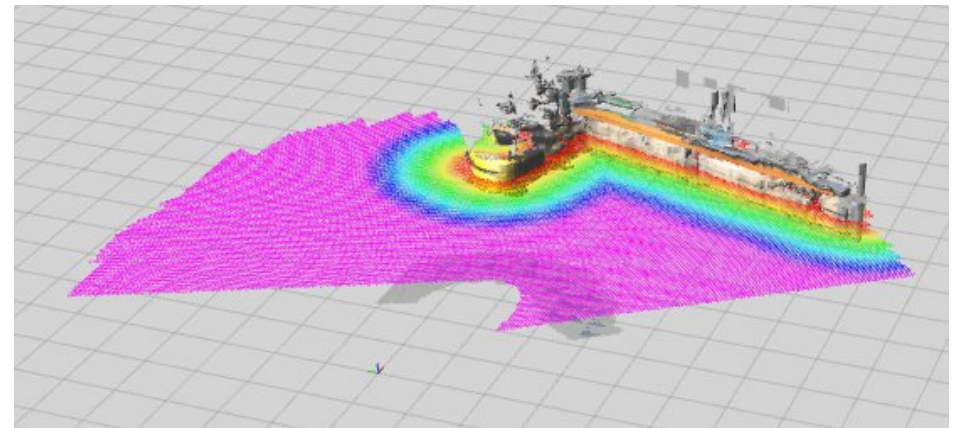


▶ OA

Ros2, Linux, Pytorch

▶ Language

Python



PROJECTS

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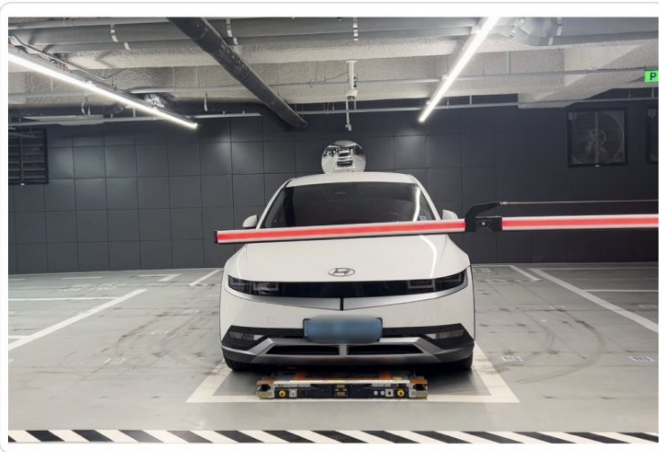
PROJECTS

Designing Connection Scenarios for Robot-Friendly Buildings and Enhancing Business Models 🛠️

Funded by Hyundai NGV (Affiliated with Hyundai Motors)

Project Timeline: 2024.06.05 – 2025.01.31

Developed standards and certification procedures for robot-friendly buildings, optimizing services through simulations.



Hyundai Wia Automated Parking Robot



Hyundai Wia Food Delivery Robot

▶ Key Role:

Intellectual Property
Strategist, Patent Analyst

▶ Key Contributions:

Simulation Analysis: Conducted metrics-based analysis to optimize robot-friendly environments and improve operational efficiency.

Standard Development: Designed protocols for autonomous information exchange, set tailored support criteria, and proposed sustainable growth solutions.

Patent Application: Reviewed competitor patents, analyzed key components, and contributed to system optimization and patent filings.

▶ Tools Used:

OA: Microsoft Excel

Engineering Tools: Confidential simulation framework

PROJECTS

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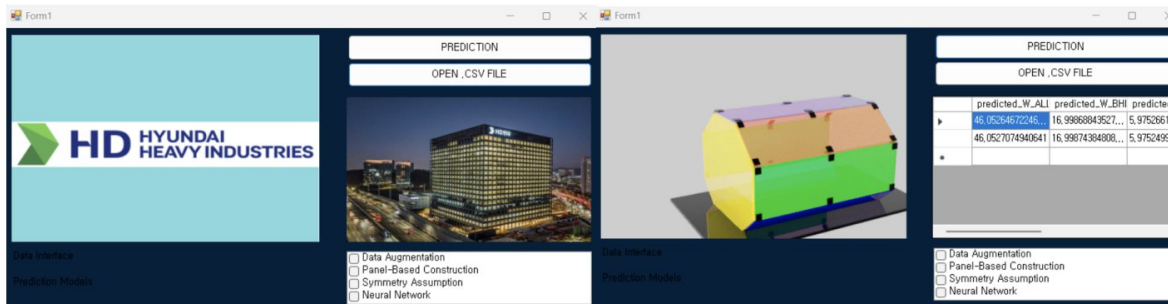
Position-specific sloshing load prediction program using deep learning

Funded by Hyundai Heavy Machinery

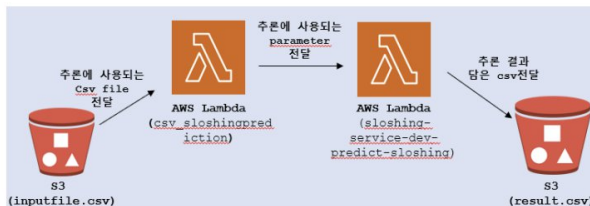
Project Timeline: 2023.03.01-2023.11.30

Responsible for building a login feature using AWS implementing functionality for administrators to upload AI models, and visualizing real-time AI model prediction results

Software Overview



Project Workflow



▶ Key Role:

Ros2, Linux, Pytorch

▶ Key Contributions:

Python

▶ Key Contributions:

Python

▶ Tools Used:

OA : AWS, Visual Studio .NET, Pytorch
Language : C#, Python

PROJECTS

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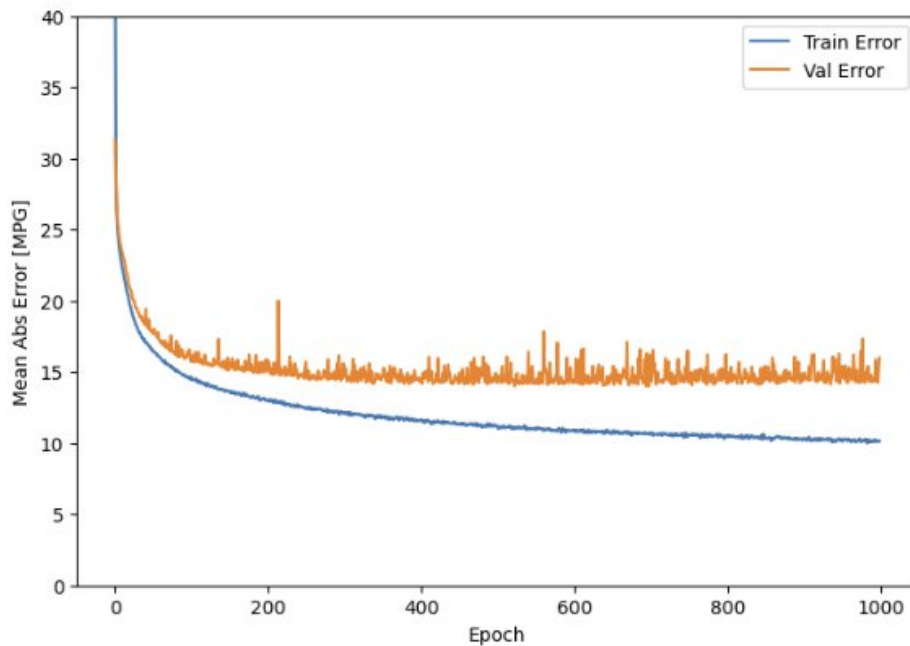
PROJECTS

Development of Small-Scale Marine Mobility for Collecting and Recycling Coastal Floating Waste 🚢

Funded by Ministry of Science and ICT

Project Timeline: 2023.03.01-2024.02.29

Scaled ship movement time-series data and enhanced AI performance using RNN, LSTM, and GRU models



▶ OA

AWS, Visual Studio .NET, Pytorch

▶ Language

C++/C#, Python

Potential Research

About me

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🔍 Research Interest – Synthetic & Multimodal Learning 🏠

Background

- Built synthetic image & segmentation datasets in NVIDIA Omniverse
- Fine-tuned detection/segmentation models for autonomous navigation
- Learned synthetic data = causal control for variation & generalization

Inspiration from Prof. Singh's Work

- **Synthetic Audio Generation:**

Used carefully controlled audio pairs instead of simple augmentations

Helps AI learn meaningful distinctions between sounds

Resonates with my work on parameterized variations in Omniverse

- **Multimodal Dataset Audits:**

Identified hidden licensing issues and data imbalances in text, speech, and video

Highlights challenges I faced ensuring my synthetic datasets were reliable and fair